Lab 6 – Code and Branch Coverage Report

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Section G

Forked repo link :

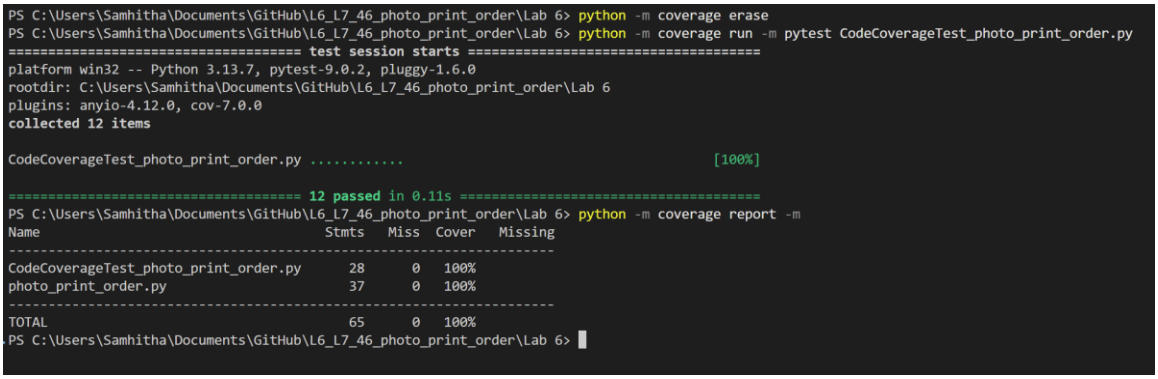
https://github.com/Samhitha1504-007/L6_L7_46_photo_print_order/tree/main/Lab%206

Overview

This lab tests `photo_print_order.py`, a module that works out extra-print charges, checks whether an order can be accepted and reports dispatch status. The starter tests covered only 38% of the code, so two new test files were written to bring line and branch coverage above 90%. All commands were run with `python -m coverage`, since Windows blocked `pip.exe` directly.

Code coverage

`CodeCoverageTest_photo_print_order.py` has 12 tests, written so every line runs at least once. They cover the error checks, the member bonus, matte and canvas pricing, all paths of `order_valid`, and each dispatch result. Line coverage reached 100%.

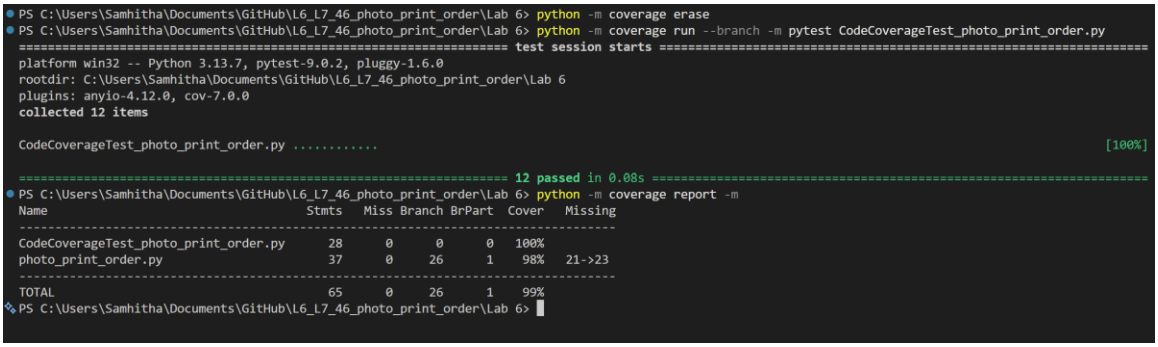


```
PS C:\Users\Samhitha\Documents\GitHub\L6_L7_46_photo_print_order\Lab 6> python -m coverage erase
PS C:\Users\Samhitha\Documents\GitHub\L6_L7_46_photo_print_order\Lab 6> python -m coverage run -m pytest CodeCoverageTest_photo_print_order.py
==================================== test session starts ====================================
platform win32 -- Python 3.13.7, pytest-9.0.2, pluggy-1.6.0
rootdir: C:\Users\Samhitha\Documents\GitHub\L6_L7_46_photo_print_order\Lab 6
plugins: anyio-4.12.0, cov-7.0.0
collected 12 items

CodeCoverageTest_photo_print_order.py ............                                     [100%]

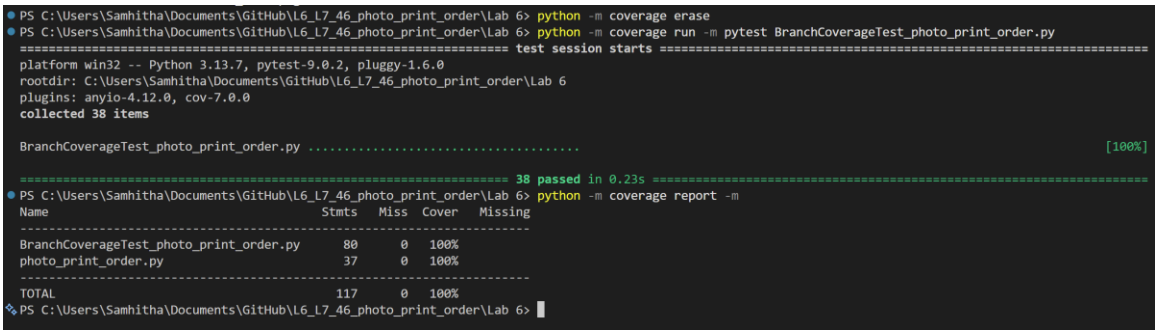
==================================== 12 passed in 0.11s =====================================
PS C:\Users\Samhitha\Documents\GitHub\L6_L7_46_photo_print_order\Lab 6> python -m coverage report -m
Name                                  Stmts   Miss  Cover   Missing
-------------------------------------------------------------------
CodeCoverageTest_photo_print_order.py      28      0   100%
photo_print_order.py                       37      0   100%
-------------------------------------------------------------------
TOTAL                                      65      0   100%
PS C:\Users\Samhitha\Documents\GitHub\L6_L7_46_photo_print_order\Lab 6>
```

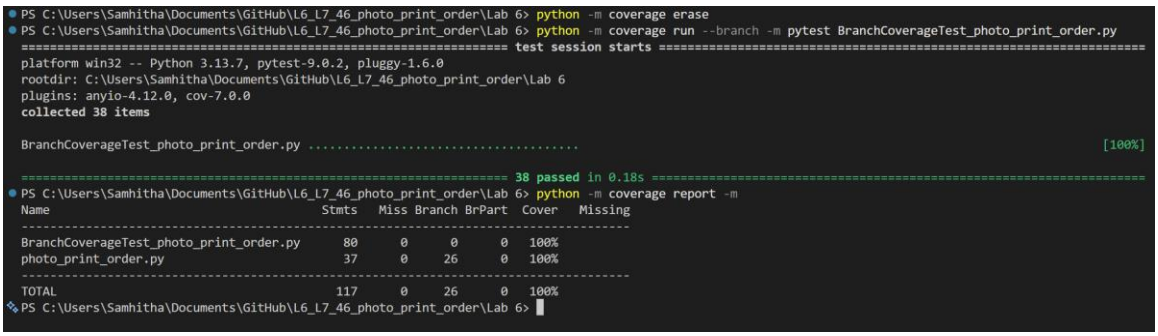
With `--branch`, coverage dropped to 98%, with branch `21->23` missing. Line 21 checks `if finish == "canvas"` inside the bulk-order block. The only bulk test used canvas, so that check was never False. Every line ran, but one path was never tested.



Branch coverage

`BranchCoverageTest_photo_print_order.py` adds a bulk glossy order to cover the missing branch, and a member ordering 8x10 prints, who gets no bonus. It also adds boundary tests: exactly 0 and 1 extra copies, 50 and 51 extra, a 25 MB file, 50 and 51 rush copies, 500 and 501 copies, and each dispatch cutoff exactly. All 38 tests passed, with 100% line and 100% branch coverage.





Conclusion

100% line coverage can still miss whole paths through the code. Branch coverage caught the gap, and the boundary tests check the comparisons where off-by-one mistakes usually happen. The two test files were committed separately, each with its screenshots.